	$L = \{ a^n b^m a^{n+m} \mid n, m >= 0 \}$	3	4

B. Tech. (Computer Engineering) Vth Semester 2nd Sessional 2025-26, AUTOMATA THEORY (CSC-505 / CEN-502) Time: 1 Hour MM:12 Note: Attempt all questions. 23BCS060 Date: 12/11/25 Your Roll No :	Q No.	CO Mapped
	1	3
	2	3
	3	4

Q. No	Question	Marks
1	Construct the CFG for the following language with suitable verifications : $L1 = \{a^i b^j c^k \mid$ either $i=j$ or $j=k$ and $i, j, k >= 1\}$. ***Validate your answers using suitable examples.***	2
2	Convert following CFG to CNF $S \rightarrow SaAB \mid AaBa$ $A \rightarrow CB \mid a$ $B \rightarrow CD \mid b$ $C \rightarrow cD \mid \varepsilon$ $D \rightarrow \varepsilon$ ***Mention all the steps and the corresponding rules used at appropriate positions.***	6
3	Construct a PDA for the followings: i) $L1 = \{ b(a^* + b^*)a^* \}$. ii) $L2 = \{ wcw^R, \quad w \in \{a, b\}^* \}$. ***Validate your answers using suitable examples.***	4